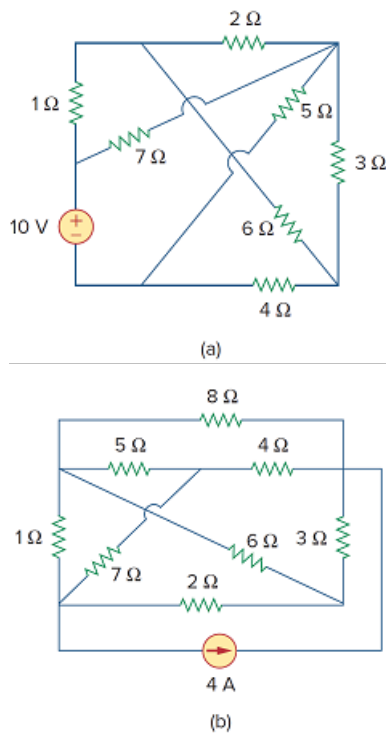


Problem

Determine which of the circuits in Fig. 3.83 is planar and redraw it with no crossing branches.

Figure 3.83:



Step-by-step solution

Step 1 of 3

The circuit that can be drawn in a plane with no branches crossing one another is called planar circuit.

The circuit other than planar circuit is called Non-Planar.

(a)

Refer to the circuit diagram in Figure 3.83(a) in the textbook.

Redraw the circuit for node identification.

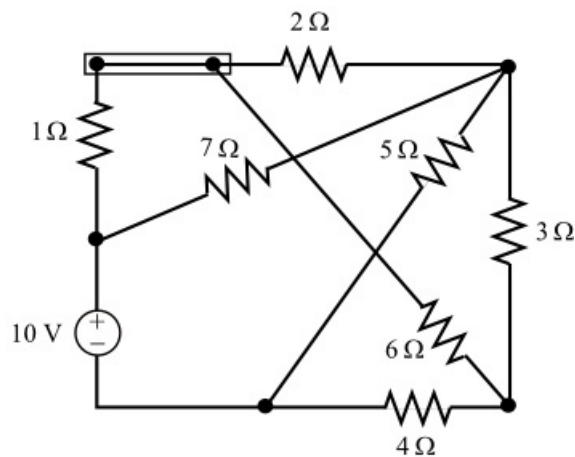


Figure 1

Step 2 of 3

Try to draw Figure 1, such that it has no crossing branches.

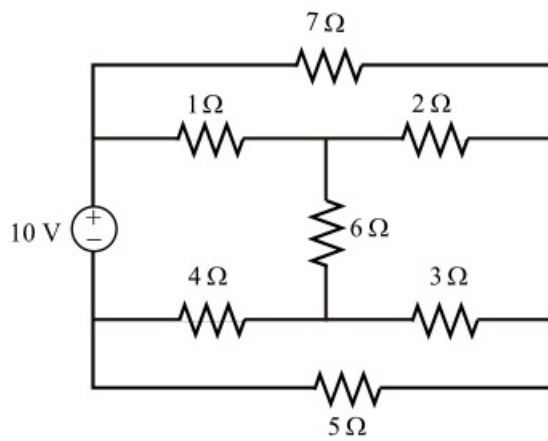


Figure 2

From Figure 2, it is possible to draw Figure 1 without crossing branches. So, the circuit is **planar**.

Step 3 of 3

(b)

Refer to the circuit diagram Figure 3.83(b) in the textbook.

Try to draw Figure 3.83(b), such that it has no crossing branches.

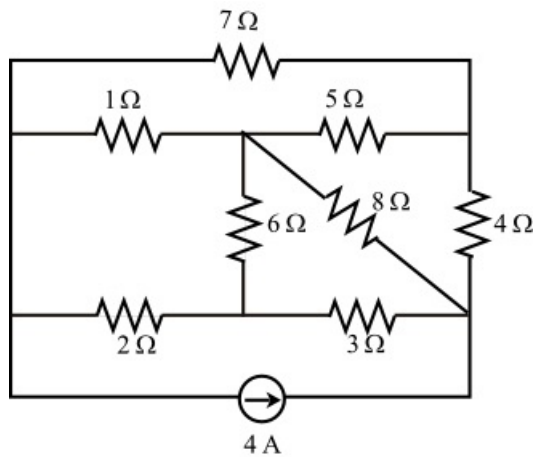


Figure 3

From Figure 3, it is possible to draw Figure 3.83(b), without crossing branches. So, the circuit is **planar**.